



CHAPTER

11

Purification and Analysis of Organic Compounds

Sr. No.	Concept	Description	Other information
1.	Methods of Purification of Organic Compounds		
	(a) Crystallization	Based on the difference in the solubilities of the compound & the impurities in a suitable solvent. e.g. - Purification of sugar, purification of potash alum	
	(b) Sublimation	Solid converted → vapour e.g.- Purification of iodine, Camphor, Naphthalene, ammonium chloride	
	(c) Distillation	Based on difference in b.p. of components of mixture. Separate volatile liquids from nonvolatile impurities.	
	Types of distillation		
	(i) Simple distillation	To separate liquids which have sufficient difference in b.p. e.g.- chloroform (334K) & aniline (457K)	
	(ii) Fractional distillation	To separate liquids which have small difference in b.p. e.g.- To separate different fractions of crude oil in petroleum industry.	
	(iii) Vacuum distillation (distillation under reduced pressure)	This method is used to purify liquids having very high boiling points but they decompose at or below their boiling points. e.g.- Glycerol is separated from spent -lye in soap industry.	
	(iv) Steam distillation	To separate substances which are steam volatile & are immiscible with water. e.g. - Aniline, Nitrobenzene, o- nitro phenol.	

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	(d) Differential extraction (e) Chromatography Types of Chromatography (i) Adsorption Chromatography - Column Chromatography - Thin layer chromatography (TLC) (ii) Partition chromatography (liquid paper chromatography)	Based on difference in solubility of organic compounds in different solvents. e.g. Phenol can be extracted using sodium hydroxide. Chromatography is used- - To purify compound - To separate mix - To test purity of compound Phases of Chromatography- - Stationary phase – e.g. Solid/liquid - Mobile phase – e.g. Pure solvent or mix. of solvents or gases Based on the fact that different components of a mixture have different degrees of adsorption on adsorbent (silica gel or alumina) Stationary phase - solid Mobile phase - liquid/gas Involves separation of a mixture over a column of adsorbent (stationary phase) packed in a glass tube. Involves separation of substances of a mixture over thin layer of an adsorbent coated on glass plate. Retention factor: Express degree of adsorption of each Component of mixture. $R_f = \frac{\text{Distance moved by component of mixture from base line}}{\text{distance moved by solvent (mobile phase) from base line.}}$ Based on continuous differential partitioning of components of a mixture between stationary & mobile phase. Stationary phase – liquid Mobile phase – liquid	
2.	Qualitative analysis of organic compound	In addition to carbon and hydrogen, organic compounds may also contain oxygen, nitrogen, Sulphur, halogens and phosphorus.	

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	(a) Carbon	$2\text{CuO} + \text{C} \xrightarrow{\Delta} 2\text{Cu} + \text{CO}_2 \uparrow$	CO ₂ gas turns lime water milky.
	(b) Hydrogen	$\text{CO}_2 \uparrow + \underset{\text{Lime water}}{\text{Ca(OH)}_2} \longrightarrow \underset{\text{Milky}}{\text{CaCO}_3} \downarrow + \text{H}_2\text{O}$ $\text{CuO} + 2\text{H} \xrightarrow{\Delta} \text{Cu} + \text{H}_2\text{O}$ $\underset{\text{White}}{\text{CuSO}_4} + 5\text{H}_2\text{O} \longrightarrow \underset{\text{Blue}}{\text{CuSO}_4 \cdot 5\text{H}_2\text{O}}$	Water droplets appear on the cooler part of the ignition tube and also turns anhydrous CuSO ₄ blue.
	(c) Nitrogen	Lassaing's extract (L.E.) $\text{Na} + \text{C} + \text{N} \xrightarrow{\Delta} \underset{\text{(L.E.)}}{\text{NaCN}}$ $\text{FeSO}_4 + 2\text{NaOH} \longrightarrow \text{Fe(OH)}_2 + \text{Na}_2\text{SO}_4$ $\text{Fe(OH)}_2 + 6\text{NaCN} \longrightarrow \text{Na}_4[\text{Fe(CN)}_6] + 2\text{NaOH}$ $\text{Na}_4[\text{Fe(CN)}_6] + \text{FeCl}_3 \xrightarrow{\text{HCl}} \underset{\text{Prussian blue}}{\text{NaFe[Fe(CN)}_6]} + 3\text{NaCl}$ or $3\text{Na}_4[\text{Fe(CN)}_6] + 4\text{FeCl}_3 \longrightarrow \underset{\text{prussian blue}}{\text{Fe}_4[\text{Fe(CN)}_6]_3} + 12\text{NaCl}$	L.E. + FeSO ₄ + NaOH, boil and cool + FeCl ₃ + conc. HCl Gives blue or green colour.
	(d) Sulphur	$2\text{Na} + \text{S} \longrightarrow \underset{\text{(L.E.)}}{\text{Na}_2\text{S}}$ $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe(CN)}_5\text{NO}] \longrightarrow \underset{\text{Sodium nitroprusside}}{\text{Na}_4[\text{Fe(CN)}_5\text{NOS}]} \quad \text{Deep violet}$ $\text{Na}_2\text{S} + (\text{CH}_3\text{COO})_2\text{Pb} \xrightarrow{\text{CH}_3\text{COOH}} \underset{\text{Black ppt.}}{\text{PbS}} \downarrow + 2\text{CH}_3\text{COONa}$	L.E. + sodium nitroprusside. A deep violet colour. L.E. + CH ₃ COOH + (CH ₃ COO) ₂ Pb Gives a black ppt.
	(e) Halogens	$\text{Na} + \text{X} \xrightarrow{\Delta} \underset{\text{(L.E.)}}{\text{NaX}}$ $\text{NaX} + \text{AgNO}_3 \xrightarrow{\text{HNO}_3} \underset{\text{ppt.}}{\text{AgX}} \downarrow$ $\underset{\text{White ppt.}}{\text{AgCl}} + 2\text{NH}_3(\text{aq.}) \longrightarrow \underset{\text{Soluble}}{[\text{Ag(NH}_3)_2]\text{Cl}}$	L.E. + HNO ₃ + AgNO ₃ . White ppt. soluble in aq. NH ₃ (or NH ₄ OH) confirms Cl. Yellow ppt. partially soluble in aq. NH ₃ (or NH ₄ OH) confirms Br. Yellow ppt. insoluble in aq. NH ₃ or (NH ₄ OH) confirms I.

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	(f) Phosphorous	$\text{P} \xrightarrow[\Delta]{\text{Na}_2\text{O}_2 \text{ boil}} \text{Na}_3\text{PO}_4$ $\text{Na}_3\text{PO}_4 + 3\text{HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + 3\text{NaNO}_3$ $\text{H}_3\text{PO}_4 + 12(\text{NH}_4)_2\text{MoO}_4 + 21\text{HNO}_3 \longrightarrow (\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$ Ammonium phosphomolybdate (yellow ppt.) $+ 21\text{NH}_4\text{NO}_3 + 12\text{H}_2\text{O}$	Solution is boiled with nitric acid and then treated with ammonium molybdate $(\text{NH}_4)_2\text{MoO}_4$. Formation of yellow ppt. indicates presence of phosphate (hence, phosphorus) in organic compound.
3.	Quantitative analysis of organic compound (a) Estimation of Carbon & Hydrogen- Liebig's combustion method (b) Estimation of Nitrogen (i) Dumas method (ii) Kjeldahl's method (c) Estimation of Halogens- Carius method (d) Estimation of Sulphur- Carius method (e) Estimation of Phosphorus-Ignition method	$\% \text{ of C} = \frac{12}{44} \times \frac{\text{mass of CO}_2 \text{ formed}}{\text{mass of compound taken}} \times 100$ $\%,$ $\% \text{ of H} = \frac{2}{18} \times \frac{\text{mass of H}_2\text{O formed}}{\text{mass of compound taken}} \times 100$ $\% \text{ of N} = \frac{28}{22400} \times \frac{\text{vol. of N}_2 \text{ at STP}}{\text{mass of compound taken}} \times 100$ $\% \text{ of N} = \frac{1.4 \times \text{molarity of acid} \times \text{vol. of acid used} \times \text{basicity of acid}}{\text{mass of compound taken}} \times 100$ $\% \text{ of X} = \frac{\text{At. mass of X}}{108 + \text{At. mass of X}} \times \frac{\text{mass of AgX formed}}{\text{mass of compound taken}} \times 100$ $\% \text{ of S} = \frac{32}{233} \times \frac{\text{mass of BaSO}_4 \text{ formed}}{\text{mass of compound taken}} \times 100$ $\% \text{ of P} = \frac{62}{222} \times \frac{\text{mass of Mg}_2\text{P}_2\text{O}_7 \text{ formed}}{\text{mass of compound taken}} \times 100$	